Integration Team Final Report



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Abstract

This report presents an insight into the work conducted this year. The fall semester was aimed at developing a method to project the electric vehicle (EV) load profile for Ottawa by 2050. Throughout the year work was done to build a optimization model to identify the optimal mix of energy generation and storage technologies required to meet projected demand. Initiated with defining project goals and forming dedicated sub-teams, the project's initial phase involved research to estimate future EV load profiles, taking into account local driving habits, temperature effects, and charging behaviors. The winter semester focused on creating a Mixed Integer Linear Programming (MILP) model in MATLAB, which considers sub-team energy sources including wind, solar, nuclear, and storage solutions, along with the potential for energy imports, to ensure a reliable and cost-effective energy supply. The model also explores the implications of different scenarios such as varying levels of decarbonization. Our findings underscore the significant impact of electrification on the power grid and highlight the critical need for strategic planning in energy generation and storage to accommodate future demands while striving for sustainability goals. The model's outcomes provide valuable insights into the feasibility of achieving a decarbonized energy system and the economic implications of different energy mixes.

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1 Introduction

The purpose of this document is to outline the work I have conducted as a member of the integration team throughout the year for the SESP capstone project. The year began with considerable time spent defining the project goals, deciding on sub-teams, and determining the best course of action to achieve this year's goals.

During the Fall semester, work was focused on researching the best way to use empirical data to project the 2050 electric vehicle (EV) load profile. This load profile is a key consideration when trying to project Ottawa's electricity demand for 2050. This work was finished during the fall semester, and the methodology and assumptions made are detailed in [1]. However, the methods and results of this work are summarized in section 2 of this report.

During the winter semester, the majority of work was spent building an optimization model. This model serves to find the optimal portfolio of sub-team energy generation and storage technologies to meet the projected needs of Ottawa in 2050. The model, built-in Matlab optimizes to find the cheapest system cost while meeting energy balance constraints at each time step. This is detailed in section 3 of this report.

2 2050 EV Projections

This section introduces the methodologies and assumptions guiding the projection of the 2050 EV load profile. An error was found in my Matlab script when projecting the 2050 EV load, since then

this has been fixed, and the updated energy demand is provided in this section of the report.

2.1 Current EV Charging Profiles

Currently, three EV charging profiles are being used: residential, workplace, and weekend. The residen tial and weekend profiles are based on data collected in Ottawa, while the workplace charging profiles originate from data collected in San Diego. Research suggests that 80% − 90% of consumers primarily use residential charging [2]. We have assumed that during holidays, all EVs will follow a weekend profile. Further research may explore how driving habits change on holidays such as Christmas.

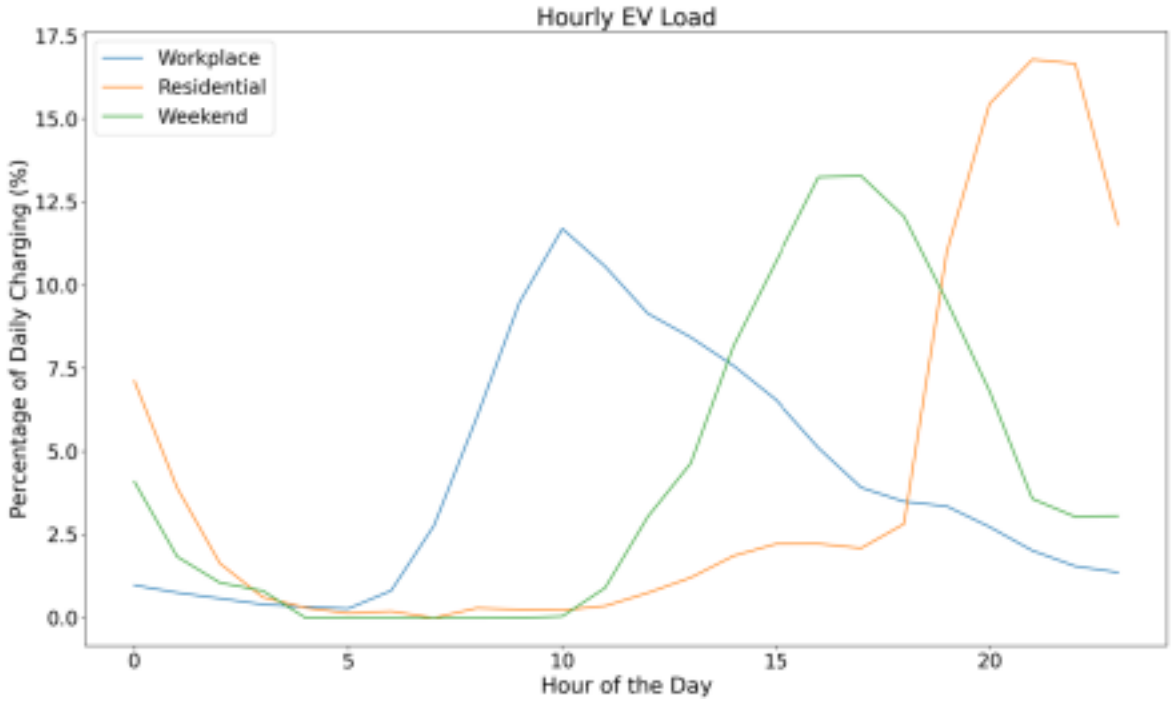


Figure 1: Charging Profiles

2.2 Total Number of EVs

To project the total number of EVs in 2050, several assumptions were necessary. First, population growth was considered to estimate the total number of EVs [3]. The number of new vehicles on the road each year was projected based on the increase in population and the number of vehicles per capita (0.486), derived from 2013 data for Ottawa [4]. Furthermore, the average vehicle lifespan in Canada is assumed to be 15.4 years [5]. The number of new EVs each year was calculated using the projected S curve for EV market share in Ontario and Ottawa's plan for 100% of new car sales to be net zero by 2040 [6]. The projected number of EVs on the road in 2050 was found to be 412, 000.

2.3 Temperature Effects

To determine the effect of temperature on EV energy consumption, we followed the trend outlined in a study that modeled the Nissan Leaf's energy consumption across different temperatures [7]. Various studies validated this model, indicating a maximum increase in energy consumption of up to 50% in cold conditions for different EV models [8],[9]. The energy consumption used in our projections is a function of the hourly temperature, assuming all vehicles follow the same relationship. Hourly temperature data was adjusted to project the impacts of climate change, forecasting a 2-degree median temperature increase by 2050 [10],[11]. The relationship between energy consumption and temperature used is shown in figure 2.

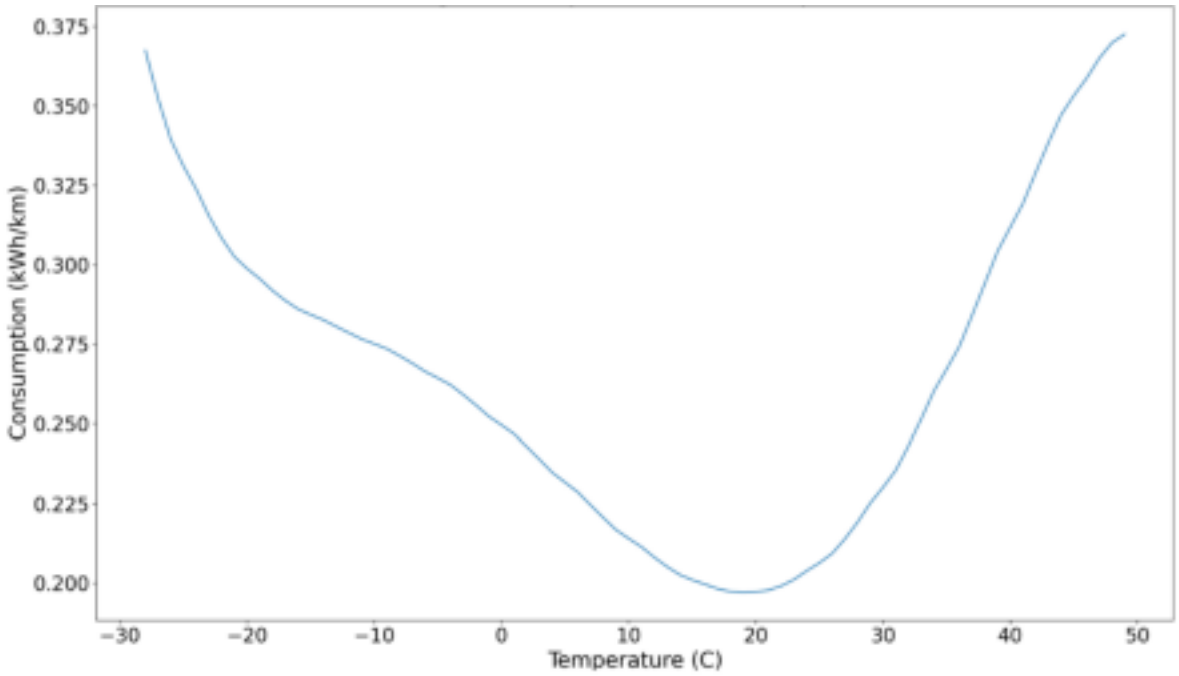


Figure 2: EV energy consumption - temperature relationship

2.4 Ottawa Commute Distances

To obtain accurate driving distance data for Ottawa, a 2011 origin-destination survey [12] was utilized. This data was used to calculate the percentile breakdown of daily driving distances in Ottawa. QGIS was used to find the centroids of each sub-area and the corresponding coordinates. A Python script utilizing the Google distance matrix API was written to find the distance by road between each sub area. The distance of travel within sub-areas was estimated using the square root of each sub-area. Using the number of trips from each origin to each destination and the corresponding distances, a MATLAB script was used to find a percentile breakdown for the distance of trips in Ottawa (figure 3). Note that this is the distance per trip, the Ottawa Transit survey outlines that on average a vehicle makes 2.67 trips per day.

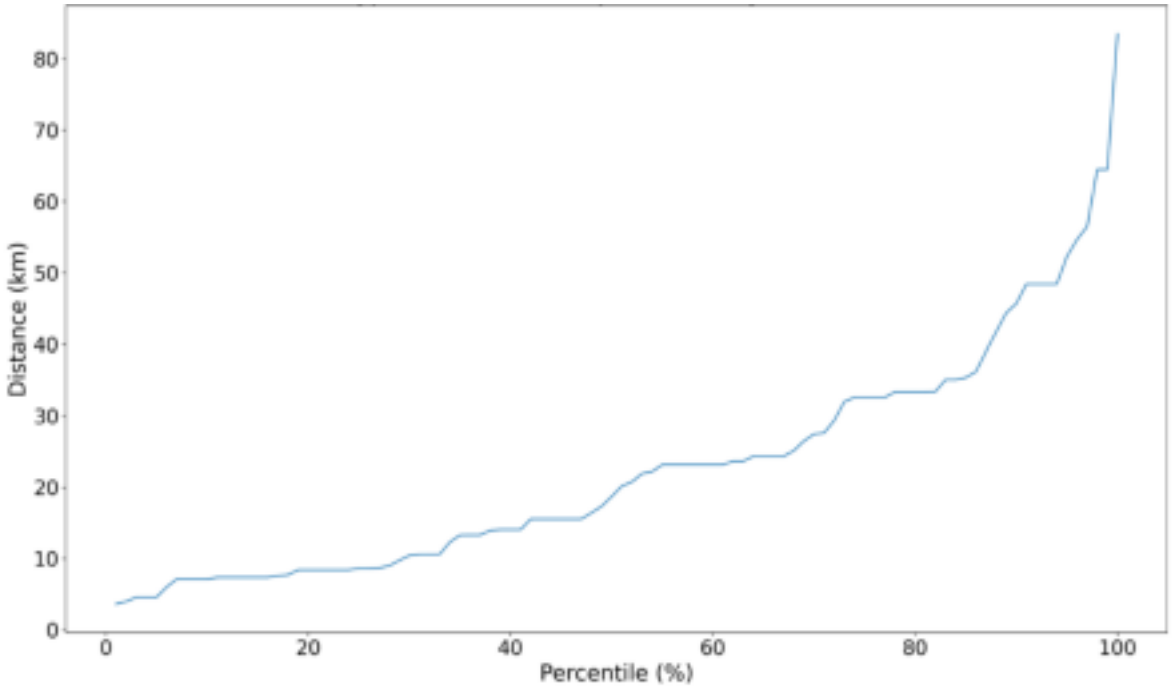


Figure 3: Commuter distance per trip

2.5 MATLAB Script

The MATLAB script imports all inputs as CSV files and prompts the user to enter the number of EVs in the community, the percentage of workplace charging, and the maximum standard deviation in hours for workplace, residential, and weekend charges. The script generates an hourly demand profile for one year and outputs it as a CSV. Figure 4 shwos the daily EV energy demand for 2050 assuming 412, 000 EVs, 15% workplace charging, and a maximum standard deviation of 2 hours.

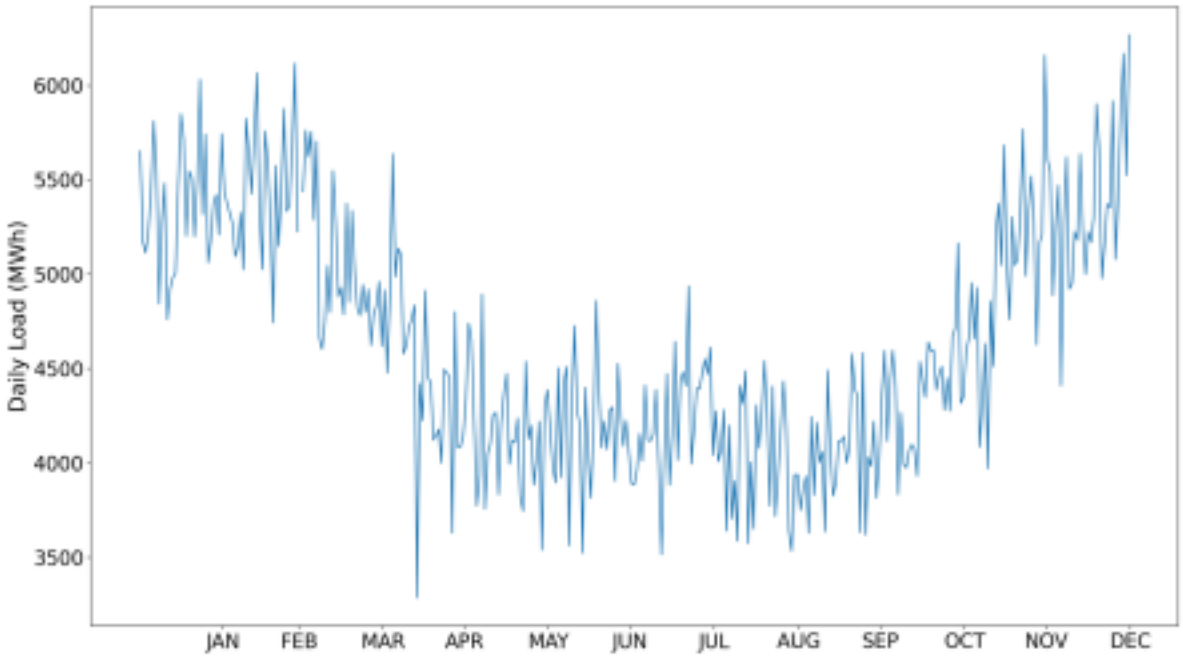


Figure 4: Daily EV Energy Demand

3 Optimization Model

It was decided that the integration team would build an optimization model to determine the optimal portfolio of energy generation and storage technologies, to meet the projected demand. The model optimizes to find the solution with the cheapest system cost. The model takes into account the lifetime of each technology being explored, and the O&M, and capital investment costs associated. When imports are being allowed the model takes into account the cost of buying imports, as well as the carbon costs associated with the current Ontario grid. A mixed integer linear programming model was built in Matlab for this purpose. The model outputs the annual cost and carbon associated with the optimal solution. It also outputs the installed capacity and hourly output of each technology. The model maintains an energy balance that ensures the hourly load is being met. We utilize parallel processing to run different scenarios at once, greatly expediting the speed of conducting sensitivity analysis. Currently, the Scenarios being run allow for different percentages of allowable imports, different carbon tax, and social cost of carbon scenarios, as well as different load reduction scenarios.

3.1 Objective Function

The model's objective function optimizes to find the cheapest annual cost. The Total Annual Cost is calculated by equation 1:

$$\text{Total Annual Cost} = \sum (\text{O\&M}) + \sum (\text{A}) + \text{Cost of Imports} \quad (1)$$

The cost of imports is the cost associated with purchasing the energy, as well as the carbon cost associated with the current Ontario grid.

The amortized annual cost of each technology is used, considering a system lifetime of 50 years.

$$A = P \cdot r(1 + r)^n \quad (1 + r)^n - 1 \quad (2)$$

where:

- A is the payment amount per annum.
- P is the capital cost.
- r is the interest rate 5%.
- n is the system lifetime (50 years).

However, since the different technologies being considered have different lifetimes the discount factor was used for technologies with lifetimes less than 50 years. The Discount Factor can be determined by
the formula:

$$\text{Discount Factor} = \frac{P}{(1 + r)^t} \quad (3)$$

where:

- P = The capital cost ($). • t = Lifetime (years).
- r = Interest rate 5

3.2 Sub-team Integration

The optimization model build needs to be capable of modeling the complex energy generation and storage technologies being considered by the various subteams. A physics-based approach was taken to model renewable energy (wind and solar). For this, the model uses hourly availability of the

resources to determine their availability. This aims to reflect the daily and seasonal variations.

Each sub-team is incorporated into our model in a manner that best represents each team's work and technology. Constraints were provided by each sub-team. When data was not obtained from the sub-team the integration team conducted research, so that model represents the technologies to the best of our ability.

3.2.1 Wind

The wind team spent the year researching and developing a novel buoyant airborne turbine (BAT). The BAT is incorporated into the optimization model based on the constraints provided by the wind team.

The optimization variable for the wind team is the number of turbines. The wind team has determined the maximum number of BATs that would be viable in Ottawa to be 100. The wind output as a function of time, t, is given by the following equation:

$$\text{windOutput}(t) = 0.5 \cdot \rho \cdot A \cdot C_p \cdot V(t)^3 \cdot N \quad (4)$$

where:

- V = Wind speed (m/s)
- N = Number of turbines
- C_p = The coefficient of performance
- A = Area of the turbine (m^2)
- ρ = Air density (kg/(m^3)

The hourly wind speeds were provided by the wind team. These values were multiplied by 1.6 to take into account the venturi effect of the BAT is shown below. The adjusted hourly windspeeds are shown below.

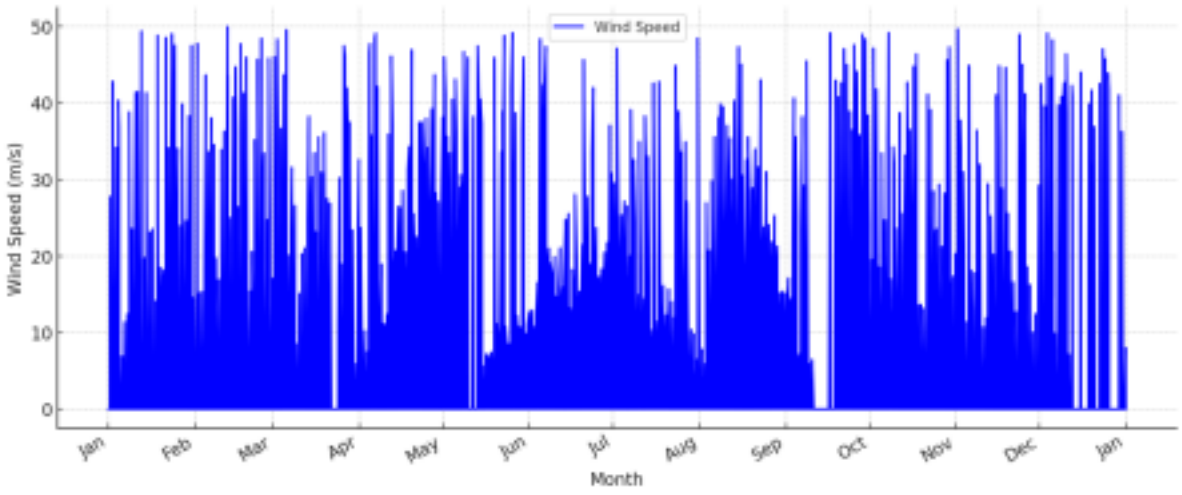


Figure 5: Hourly windspeeds

It should be noted that employing these values introduces certain challenges. The model lacks operational constraints for the BAT, allowing wind energy production to continue even at maximum wind speeds (50 m/s). In reality, however, the BAT would likely be offline at such wind speeds. This should be fixed in future versions of the model. Nonetheless, the mean wind speed of 13 m/s aligns with the expectations of the wind team for their operational altitude of 300 meters.

3.2.2 Solar

The solar output is a function of the installed area, and the hourly irradiance/temperature. The maximum area constraint was provided by the solar team and found to be 418 km2, this included all usable rooftops, parking, and agricultural land. The cost for solar being used is $363/m2, which was derived using NREL's 2050 Commercial Solar Class 10 Moderate cost assumption of $1690/kW, and the Solar Team's panel rating and area of 550W and 2.56m2, respectively [13].

The solar output as a function of time, t, is given by equation 5:

$$\text{solarOutput}(t) = \eta_p(t) \cdot \text{solarPanelArea} \cdot \text{irradiance}(t) \quad (5)$$

The efficiency parameter, denoted by η_p, is defined as:

$$\eta_p(t) = 21.51 - 0.0034\, T_{\text{amb}}(t) + \frac{0.8}{45-20} \cdot 0.5 - 25 \quad (6)$$

where:

- η_p = Solar panel efficiency.
- $T_{amb}(t)$ = Ambient temperature (Celsius)

The hourly irradiance shown below was provided by the solar team.

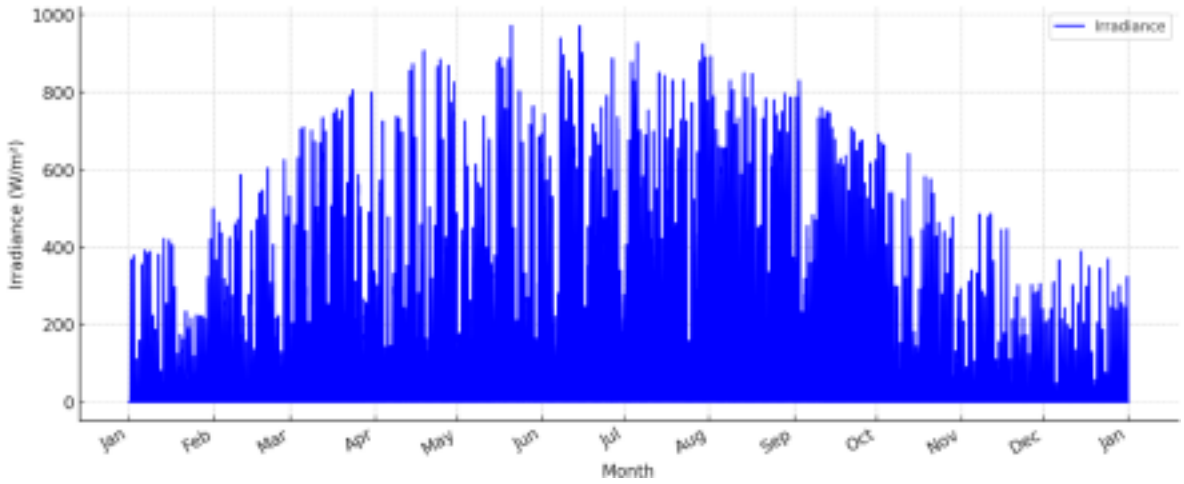


Figure 6: Hourly irradiance

3.2.3 Nuclear

The Nuclear output in our model is constrained by the maximum ramp rate and the number of reactors. This was determined to be 5% of reactor capacity [14]. The model considers a maximum of 7, 300MW reactors. With an overnight capital cost and operation and maintenance cost of $24.0 Million/MW and $10/MWh. These were based on the National Renewable Energy Laboratory

(NREL) 2050 Conservative SMR cost estimate [13.]

3.2.4 Energy Storage

Storage follows the constraints as outlined by the storage team for the proposed HESS system using a 20:1 Li-ion-flywheel split. Binary variables are used to ensure the energy storage system can only be charging or discharging at any given time. When storage is charging is considered an additional load by the model. Additionally, the state of charge of the storage system is monitored at each timestep to ensure it adheres to the minimum and maximum capacity limits. Finally, the model ensures that the initial and final SOC of storage are equal.

Table 1: Combined HESS

Variable	Units	Value
Minimum SoC	%	20
Charge Rate	MW	187
Discharge Rate	MW	187
Initial SoC	%	20
Initial Capital Cost	$	297,835,000
Operational Costs	$	1,579,100
Round Trip Efficiency	%	94
Maximum Capacity	MWh	713

The model assumed a Maximum possible capacity of 2100 MWh or 3 of the proposed facilities.

3.2.5 Sustainable Buildings

The sustainable building team supplied a residential load profile encompassing three scenarios: heat pumps only, limited retrofits, and an unlimited budget profile. These profiles serve as inputs for the optimization model. The model was executed for each scenario to examine the effects of load reduction.

It should be noted here that the commercial load profile is related to the residential load and was adjusted accordingly for each scenario.

3.3 Energy Imports

Energy imports are also included in this model. The initial goal of this project was to be able to achieve total decarbonization (zero imports allowed). When running the model, it often was not solvable for zero imports.

When including energy imports data was taken directly from the Independent Electricity System Operator (IESO). This was to find the cost to buy this energy $/MW as well as the carbon associated with this energy. It was determined that 8.5% of energy was produced using natural gas. As a result, the model uses a fixed cost for energy imports of $24/MW and a fixed carbon emission associated with energy imports of 0.021tonnes/MW [13],[15].

The model was run for different carbon tax, and social cost of carbon scenarios, ranging from $315/tonne - $920/tonne [16],[17]. However, because of the minimal carbon currently associated with the Ontario grid, this had no reflection on the solution.

4 Model Results

The model was run for different scenarios of load reduction. It was also run for different allowable import percentages. The total load for these three scenarios is shown below.

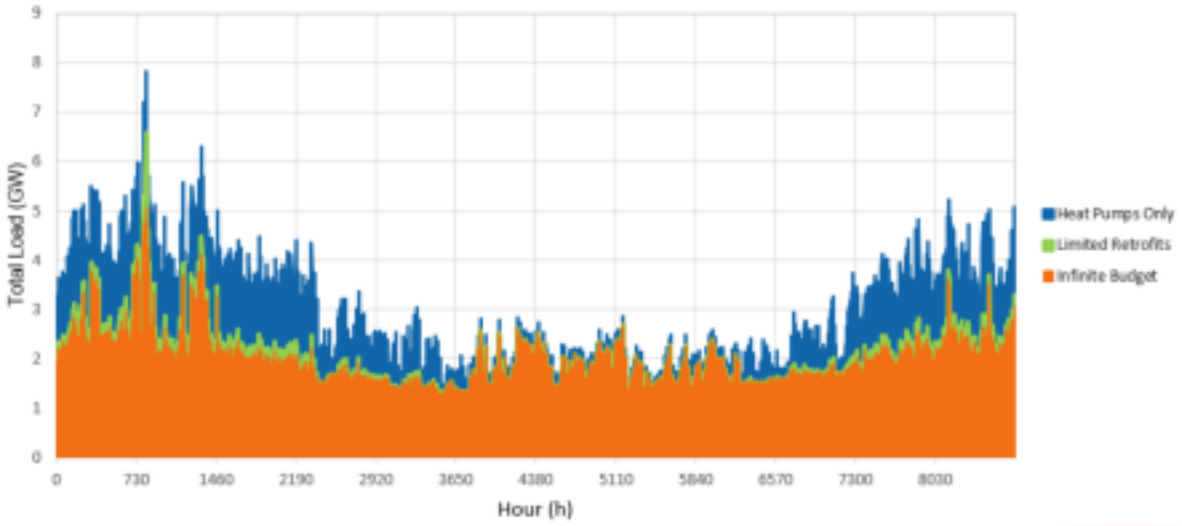


Figure 7: Total loads

Four representative weeks from each season were taken to hasten the model runtime. An example of the stacked chart illustrating the hourly output from each technology is shown in the figure below.

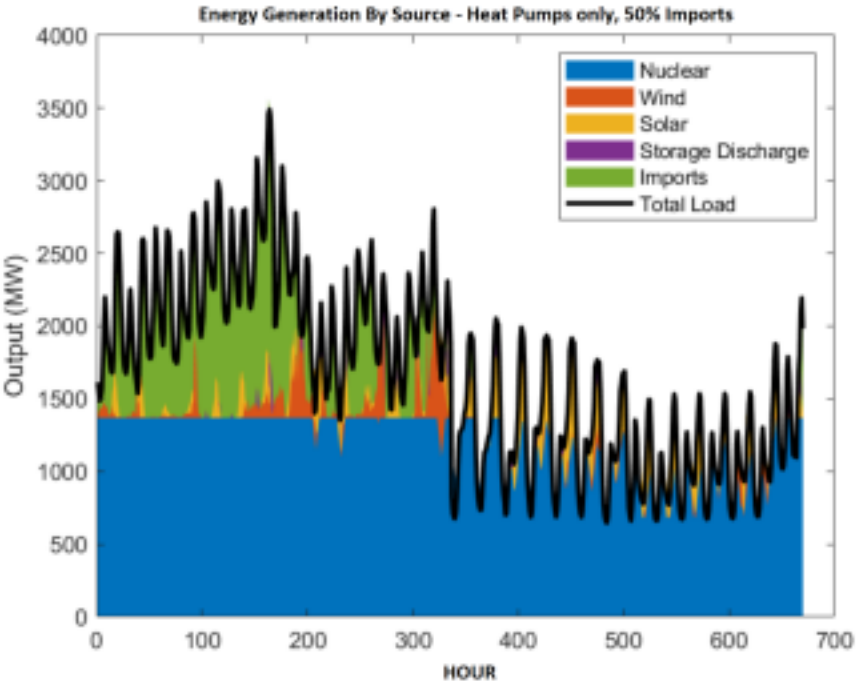


Figure 8: Model Output

The results below show the installed capacity of each technology for the various scenarios that were run.

Scenario	Metric	Allowable Import Percentage			
		0%	25%	50%	100%
Heat Pumps Only	Nuclear (MW)	N/A	N/A	2,100	0
	Wind (MW)	N/A	N/A	880	0
	Solar (MW)	N/A	N/A	380	0
	Storage (MWh)	N/A	N/A	710	0
Limited Retrofits	Nuclear (MW)	2,100	2,100	1,200	0
	Wind (MW)	430	640	840	0
	Solar (MW)	27,500	230	0	0
	Storage (MWh)	1,420	710	710	0
Infinite Budget	Nuclear (MW)	2,100	1,500	1,200	0
	Wind (MW)	380	490	370	0
	Solar (MW)	4,960	0	0	0
	Storage (MWh)	1,420	710	710	0

Table 2: Model Results: Installed Capacity

Furthermore, the table below shows the annual cost and annual carbon associated with each solu tion. The general trend for increased cost as decarbonization is achieved can be seen.

Scenario	Metric	Allowable Import Percentage			
		0%	25%	50%	100%
Heat Pumps Only	Cost ($Billion)	N/A	N/A	16.9	14.5
	Carbon (MT CO_2e)	N/A	N/A	67.3	390.1
Limited Retrofits	Cost ($Billion)	22.4	16.2	15.4	14.2
	Carbon (MT CO_2e)	0	11.2	64.2	253.8
Infinite Budget	Cost ($Billion)	29.3	16.1	15.4	14.2
	Carbon (MT CO_2e)	0	10.9	58.8	240.7

Table 3: Model Results: Cost

The initial goal of this project is to meet Ottawa's energy generation and storage needs while using the technologies being explored by each subteam. However, when operating within the constraints defined by the subteams, it was not solvable for the zero imports and heat pump-only scenario. In general, the trend of increased costs to achieve decarbonization can be seen.

It can also be seen that while the model solves for zero imports and limited retrofits, the solution requires 27 GW of solar. Resulting in significant amounts of curtailment.

5 Conclussion

With the increasing adoption of electric vehicles (EVs), substantial impacts on the power system are anticipated. Our projection method, which relies on empirical data, models the load specific to Ottawa by incorporating local driving habits and charging profiles, while also accounting for heightened energy demands during extreme temperatures.

Given the opportunity, exploring the impact of Level 3 chargers would be of interest. As the preference for rapid charging grows alongside EV adoption, the capabilities of Level 3 chargers, which can charge at rates up to 350 kW, become particularly relevant [18].

Applying our current methodology to estimate the energy consumption per EV in Ottawa could prove insightful. By employing the established charging profiles, we could obtain the charging start times in Ottawa. Depending on the proliferation of superchargers, an analysis of Level 2 versus Level 3 chargers could then be conducted. Level 3 chargers could either pose a risk or offer a benefit to the grid. Although such chargers would generate significant peaks in demand, they could also promote workplace charging, shifting the peak EV load to off-peak hours.

The optimization model has met the goals of the integration team outlined for the project this year. Nonetheless, there is considerable scope for enhancement.

Incorporating variable energy prices for imports and the carbon footprint of these imports would be a wise addition. Given Ontario's population growth and the rapid pace of electrification, the entire network will need significant capacity expansion. The Independent Electricity System Operator (IESO) anticipates a 22 GW shortfall in Ontario's energy supply by 2050. Our model currently adopts a uniform import rate, yet the cost of energy imports is expected to increase in the coming years. Moreover, the carbon footprint associated with energy imports is likely to vary. Integrating the fluctuation in energy prices between peak and off-peak hours could further refine the model, potentially enhancing the utility of energy storage through energy arbitrage opportunities.

Lastly, our model represents Ottawa as a single node without considering grid inertia. It may be beneficial to model Ottawa with multiple nodes and implement constraints to ensure grid stability and maintain inertia.

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